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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc Degree Examination - APRIL 2021

CH509S – ORGANIC CHEMISTRY - III

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Tertiary nitro alkanes cannot tautomerise because they
   a) Are not stable b) Are strong base
   c) Do not contain labile hydrogen d) All the above

2. Schiff's base on reduction give
   a) Aniline b) Primary amine c) Secondary amine d) Tertiary amine

3. The most stable conformer of cyclohexane is,
   a) Chair b) Boat c) Skew boat d) Twisted boat

4. How many chiral stereoisomers can be drawn for $CH_3CHFCHFCH(CH_3)_2$
   a) 1 b) 3 c) 2 d) 4

5. Malonic ester react with acetaldehyde gives
   a) Crotonic acid b) Cinnamic acid c) Succinic acid d) Laevulinic acid

6. The cyclic structure of diazomethane is highly
   a) Stable b) Unstable c) Acidic d) None of the above

7. In a Benzilic acid rearrangement
   a) Benzaldehyde is converted into benzoin b) Benzoin is converted into benzilic acid
   c) Benzilic acid is converted into benzil d) Benzil is converted into benzilic acid

8. In which of the following rearrangement amide is reduced to amine which has one carbon less than the starting material?
   a) Lossen rearrangement b) Beckmann rearrangement
   c) Diekmann rearrangement d) Hoffmann rearrangement

9. Cellulose fibers resemble with protein structure in the form of
   a) ß - sheets b) α - helices c) ß -turns d) None of these

10. Which of the following is not a disaccharide
    a) Amylase b) Cellobiose c) Lactose d) None of these

II. Fill up the blanks

11. Nitrobenzene when reduced with Sn and HCl gives _____

12. Hexane and 3-methyl pentane are example of _____

13. Diazomethane is used to prepare _____ ester from carboxylic acids.

14. Conversion of aromatic ketoximes to acid amide with PCl_5 is called _____ rearrangement.

15. Example of monosaccharide is _____

III. Match the following

16. Shape of amine	-	Building blocks of proteins
17. Boat and chair conformation	-	Primary amine
18. Active methylene group	-	Pyranose sugars
19. Hoffmann rearrangement	-	Acetoacetic ester
20. Amino acids are	-	Pyramids

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. How tertiary amines are prepared?
22. Write the preparation of diazonium salt from aniline.
23. What is optical activity?
24. Define Walden inversion.
25. What is erythro and threo form?
26. How is acetoacetic ester prepared in the laboratory?
27. Write the resonance structures of diazomethane.
28. What is keto - enol tautomerism?
29. Define primary structure of proteins.
30. What is fries rearrangement?
31. How fructose can be converted to glucose?
32. What is Zwitter ion?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Write any four properties of nitro compounds. **(4)**
    b) How 1, 3, 5-trinitrobenzene are prepared? **(3)**

34. a) Explain the elements of symmetry. **(4)**
    b) Describe the optical activity of Allenes? **(3)**

35. a) What is racemization?and explain the methods of racemization. **(4)**
    b) Discuss the different conformations of 1, 3 Disubstituted- cyclohexane? **(3)**

36. a) How is acetaldehyde converted into crotonic acid? **(4)**
    b) Write the preparation of diazoacetic ester? **(3)**

37. a) Explain the reaction and mechanism of pinacol – pinacolone rearrangement. **(5)**
    b) What is intermolecular rearrangement? **(2)**

38. a) Explain the reaction and mechanism of Baeyer -villiger rearrangement? **(5)**
    b) Write any two properties of starch. **(2)**

39. a) Explain the structural elucidation of glucose **(5)**
    b) How amino acids are synthesised by Gabriel phthalimide method? **(2)**

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